



■ LED Traffic Light

Circuits — Science Lesson

Grade	Grade 1 (Ages 5–7)
Duration	30 Minutes
Subject	Science — Circuits & Road Safety
Level	Beginner
Platform	Planrs Academy



LED Traffic Light — Position of Lights

A magical journey into circuits, LED lights, and road safety!



SECTION 1 — Learning Objectives

By the end of this lesson, students will be able to:

LO 1 — IDENTIFY	Recognise a traffic light and name all three coloured lights: Red, Yellow (Amber), and Green.
LO 2 — UNDERSTAND MEANING	State what each colour signal means: Red = Stop, Yellow = Get Ready, Green = Go.
LO 3 — CONNECT TO LIFE	Give at least one real-life example of where they have seen a traffic light in their city, town, or village.

LO 4 — UNDERSTAND LED	Explain in one simple sentence why LED lights are used (bright, easy to see, and save electricity).
LO 5 — APPLY SAFETY	Demonstrate correct crossing behaviour at a pedestrian crossing by following traffic light signals.
LO 6 — OBSERVE & DESCRIBE	Describe the order in which traffic light colours appear.

KEY VOCABULARY

TERM	MEANING
LED	Light Emitting Diode — a tiny chip that makes bright light using very little electricity.
Traffic Light	A set of coloured signal lights on a pole at road crossings to control vehicles and people.
Circuit	A closed loop through which electricity flows — from battery, through wires, through the LED, and back.
Resistor	A small part in a circuit that controls how much electricity flows so the LED does not burn out.
Pedestrian	A person who is walking, especially on or near a road.
Zebra Crossing	Black-and-white striped road markings showing where people can cross the road safely.
Junction / Chowk	A place where two or more roads meet — often where traffic lights are placed.
Signal Sequence	The order in which traffic lights change: Red → Yellow → Green → Yellow → Red.

SECTION 2 — Concept Notes (Student-Friendly)

1. What is a Traffic Light?

A **traffic light** is a set of coloured signal lights placed at road crossings, junctions, and busy streets to control the movement of vehicles and people.

■ ■ **Indian Example:** The signal post you see near your school gate or at the big road crossing in your area — like at a busy chowk in Mumbai, Bengaluru, Chennai, or Delhi.

2. The Three Colours and What They Mean

COLOUR	POSITION	MEANING	WHAT TO DO
RED	Top	STOP	Stand still. Do not cross. Wait.

YELLOW	Middle	GET READY	Slow down. Be ready for the change.
GREEN	Bottom	GO	Cross safely. Look both ways first.

3. What is an LED?

LED stands for **Light Emitting Diode**. It is a tiny electronic part that glows brightly when electricity flows through it. Traffic lights in India now use LED bulbs instead of old, regular bulbs.

Very Bright	LED lights are so bright that drivers can see them even in strong sunlight.
Save Electricity	LEDs use much less power than old bulbs — they save energy for our cities.
Last Longer	LEDs work for many years without needing to be changed, saving time and money.

4. The Order of Lights (Signal Sequence)

Traffic lights always follow the **same order**. This order never changes, so everyone knows what to expect next:



5. How Does an LED Traffic Light Circuit Work?

A circuit is a closed loop that lets electricity travel from a power source (battery), through connecting wires and a resistor, and into the LED — making it glow. When the circuit is broken (switch off), the light goes out.



Teaching Rhyme — Say It Together!

"Red says STOP — don't go yet! Yellow says WAIT — are you set? Green says GO — off you go! Traffic lights put on a show!"

SECTION 3 — Session Plan (30 Minutes)

TIME	SEGMENT	WHAT HAPPENS
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0:00–2:00	Welcome & Warm-Up	Ask: 'Have you ever stopped at a red light in an auto or car?' Activate prior knowledge. Show a colourful Indian road junction image.
2:00–5:00	Intro to Traffic Lights	Show an Indian traffic light post (cartoon or real). Introduce: 'This tells vehicles and people what to do on the road.'
5:00–10:00	The Three Colours	Teach Red, Yellow, Green one at a time with large clear visuals. Use an Indian street scene for each. Say the rhyme together.
10:00–13:00	What is LED?	Simple analogy: 'LED lights are like superheroes — super bright and they save energy!' Show old bulb vs LED side by side.
13:00–17:00	Order of Lights	Teach the sequence using animation or flip-cards. Students call out the next colour as a group activity.
17:00–21:00	Activity 1 — Circuit Build	Students drag/colour parts: Battery, Resistor, Wires, Red LED, Amber LED, Green LED onto circuit board and test.
21:00–25:00	Activity 2 — Role Play	Mini role-play: 'You are crossing the road. What do you do at Red? Yellow? Green?' Students respond with actions.
25:00–28:00	Recap & Quick Quiz	5-question rapid-fire quiz. Thumbs up / thumbs down format. Covers all Learning Objectives.
28:00–30:00	Safety Pledge & Close	Students repeat: 'Red means Stop. Yellow means Wait. Green means Go. I will always follow the traffic light!'

Indian Context References

Use these in your lesson to make examples feel real and familiar to students:

Vehicles	Auto-rickshaws, school vans, BEST / DTC / BMTC buses, bicycles, motorcycles, cars.
Cities	Mumbai, Delhi, Bengaluru, Chennai, Hyderabad, Kolkata, Pune, Jaipur, Mysuru, Nashik.
Road Terms	Chowk, flyover, zebra crossing, signal post, maidan, footpath.
Characters	Children in Indian school uniforms, traffic police officer (khaki + white gloves), auto driver.
Situations	School gate crossing in the morning, Diwali market crowd, hospital zebra crossing.

SECTION 4 — Classroom Activities

Activity A — Fill in the Blank

Complete each sentence with the correct word. Use the word box below.

Word Box: STOP WAIT GO LED Red Green Yellow circuit

1. The _____ light means STOP.

2. The _____ light means GO.
3. The Yellow light means _____.
4. _____ stands for Light Emitting Diode.
5. Electricity flows around a closed loop called a _____.
6. Traffic lights always follow this order: Red → _____ → Green.

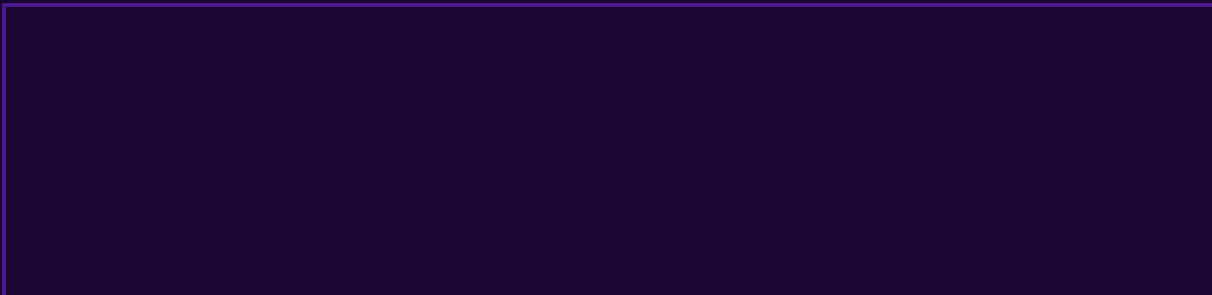
Activity B — True or False

Circle TRUE or FALSE for each statement.

STATEMENT	TRUE	FALSE
1. Red means it is safe to walk across the road.	☐	☐
2. Traffic lights always follow the same colour order.	☐	☐
3. LED lights use more electricity than old bulbs.	☐	☐
4. The Yellow light comes between Red and Green.	☐	☐
5. A zebra crossing is where people can cross the road safely.	☐	☐
6. The Green light is at the top of a traffic signal.	☐	☐

Activity C — Draw and Label the Traffic Light

Draw a traffic light in the box below. Label the three lights with their colours and meanings.



My drawing of a traffic light! 

SECTION 5 — Quick Quiz (5 Questions)

Tick (✓) the correct answer for each question.

Q1. What does the RED traffic light mean?

- A) Go
- B) Stop ✓
- C) Get Ready

Q2. Which light is at the BOTTOM of a traffic light?

- A) Red
- B) Yellow
- C) Green ✓

Q3. What does LED stand for?

- A) Light Energy Device
- B) Light Emitting Diode ✓
- C) Large Electric Display

Q4. What is the correct order of traffic lights?

- A) Green → Yellow → Red → Yellow → Green
- B) Red → Yellow → Green → Yellow → Red ✓
- C) Yellow → Red → Green → Red → Yellow

Q5. Why do traffic lights use LED bulbs?

- A) They are more colourful
- B) They are cheaper to buy
- C) They are bright and save electricity ✓

ANSWER KEY (Teacher Copy)

Fill in the Blank Answers:

1. Red | 2. Green | 3. Wait | 4. LED | 5. circuit | 6. Yellow

True or False Answers:

- 1. FALSE (Red = Stop, not safe to walk)
- 2. TRUE
- 3. FALSE (LED uses LESS electricity)
- 4. TRUE
- 5. TRUE
- 6. FALSE (Green is at the BOTTOM; Red is at the top)

Quiz Answers:

- Q1: B) Stop
- Q2: C) Green
- Q3: B) Light Emitting Diode
- Q4: B) Red → Yellow → Green → Yellow → Red
- Q5: C) They are bright and save electricity

Our Safety Pledge:

*"Red means Stop. Yellow means Wait. Green means Go.
I will always follow the traffic light!"*

